$x_n = 0$, then by excision we have $H_n(M, M - \{x\}; \mathbb{Z}) \approx H_n(\mathbb{R}^n_+, \mathbb{R}^n_+ - \{0\}; \mathbb{Z}) = 0$, whereas if x corresponds to a point $(x_1, \cdots, x_n) \in \mathbb{R}^n_+$ with $x_n > 0$ or to a point of $\mathbb{R}^n$, then $H_n(M, M - \{x\}; \mathbb{Z}) \approx H_n(\mathbb{R}^n, \mathbb{R}^n - \{0\}; \mathbb{Z}) \approx \mathbb{Z}$. Thus the points x with $H_n(M, M - \{x\}; \mathbb{Z}) = 0$ form a well-defined subspace, called the **boundary** of M and denoted ∂M. For example, $\partial \mathbb{R}^n_+ = \mathbb{R}^{n-1}$ and $\partial D^n = S^{n-1}$. It is evident that ∂M is an $(n-1)$-dimensional manifold with empty boundary.

If M is a manifold with boundary, then a **collar** neighborhood of ∂M in M is an open neighborhood homeomorphic to $\partial M \times [0, 1)$ by a homeomorphism taking ∂M to $\partial M \times \{0\}$.

Proposition 3.42. *If M is a compact manifold with boundary, then ∂M has a collar neighborhood.*

Proof: Let M' be M with an external collar attached, the quotient of the disjoint union of M and $\partial M \times [0, 1]$ in which $x \in \partial M$ is identified with $(x, 0) \in \partial M \times [0, 1]$. It will suffice to construct a homeomorphism $h : M \to M'$ since $\partial M'$ clearly has a collar neighborhood.

Since M is compact, so is the closed subspace ∂M. This implies that we can choose a finite number of continuous functions $\varphi_i : \partial M \to [0, 1]$ such that the sets $V_i = \varphi_i^{-1}(0, 1]$ form an open cover of ∂M and each V_i has closure contained in an open set $U_i \subset M$ homeomorphic to the half-space $\mathbb{R}^n_+$. After dividing each φ_i by $\sum_j \varphi_j$ we may assume $\sum_i \varphi_i = 1$.

Let $\psi_k = \varphi_1 + \cdots + \varphi_k$ and let $M_k \subset M'$ be the union of M with the points $(x, t) \in \partial M \times [0, 1]$ with $t \le \psi_k(x)$. By definition $\psi_0 = 0$ and $M_0 = M$. We construct a homeomorphism $h_k : M_{k-1} \to M_k$ as follows. The homeomorphism $U_k \approx \mathbb{R}^n_+$ gives a collar neighborhood $\partial U_k \times [-1, 0]$ of ∂U_k in U_k, with $x \in \partial U_k$ corresponding to $(x, 0) \in \partial U_k \times [-1, 0]$. Via the external collar $\partial M \times [0, 1]$ we then have an embedding $\partial U_k \times [-1, 1] \subset M'$. We define h_k to be the identity outside this $\partial U_k \times [-1, 1]$, and for $x \in \partial U_k$ we let h_k stretch the segment $\{x\} \times [-1, \psi_{k-1}(x)]$ linearly onto $\{x\} \times [-1, \psi_k(x)]$. The composition of all the h_k's then gives a homeomorphism $M \approx M'$, finishing the proof. □

More generally, collars can be constructed for the boundaries of paracompact manifolds in the same way.

A compact manifold M with boundary is defined to be R-orientable if $M - \partial M$ is R-orientable as a manifold without boundary. If $\partial M \times [0, 1)$ is a collar neighborhood of ∂M in M then $H_i(M, \partial M; R)$ is naturally isomorphic to $H_i(M - \partial M, \partial M \times (0, \varepsilon); R)$, so when M is R-orientable, Lemma 3.27 gives a relative fundamental class $[M]$ in $H_n(M, \partial M; R)$ restricting to a given orientation at each point of $M - \partial M$.

It will not be difficult to deduce the following generalization of Poincaré duality to manifolds with boundary from the version we have already proved for noncompact manifolds:

Theorem 3.43. *Suppose M is a compact R-orientable n-manifold whose boundary ∂M is decomposed as the union of two compact $(n-1)$-dimensional manifolds A and B with a common boundary $\partial A = \partial B = A \cap B$. Then cap product with a fundamental class $[M] \in H_n(M, \partial M; R)$ gives isomorphisms $D_M : H^k(M, A; R) \to H_{n-k}(M, B; R)$ for all k.*

The possibility that A, B, or $A \cap B$ is empty is not excluded. The cases $A = \varnothing$ and $B = \varnothing$ are sometimes called Lefschetz duality.

Proof: The cap product map $D_M : H^k(M, A; R) \to H_{n-k}(M, B; R)$ is defined since the existence of collar neighborhoods of $A \cap B$ in A and B and ∂M in M implies that A and B are deformation retracts of open neighborhoods U and V in M such that $U \cup V$ deformation retracts onto $A \cup B = \partial M$ and $U \cap V$ deformation retracts onto $A \cap B$.

The case $B = \varnothing$ is proved by applying Theorem 3.35 to $M - \partial M$. Via a collar neighborhood of ∂M we see that $H^k(M, \partial M; R) \approx H_c^k(M - \partial M; R)$, and there are obvious isomorphisms $H_{n-k}(M; R) \approx H_{n-k}(M - \partial M; R)$.

The general case reduces to the case $B = \varnothing$ by applying the five-lemma to the following diagram, where coefficients in R are implicit:

$$\begin{array}{ccccccccc}
\cdots \longrightarrow & H^k(M, \partial M) & \longrightarrow & H^k(M, A) & \longrightarrow & H^k(\partial M, A) & \longrightarrow & H^{k+1}(M, \partial M) & \longrightarrow \cdots \\
& \downarrow{\scriptstyle [M]\frown} & & \downarrow{\scriptstyle [M]\frown} & & \downarrow{\scriptstyle \approx} & & \downarrow{\scriptstyle [M]\frown} & \\
& & & & & H^k(B, \partial B) & & & \\
& & & & & \downarrow{\scriptstyle [B]\frown} & & & \\
\cdots \longrightarrow & H_{n-k}(M) & \longrightarrow & H_{n-k}(M, B) & \longrightarrow & H_{n-k-1}(B) & \longrightarrow & H_{n-k-1}(M) & \longrightarrow \cdots
\end{array}$$

For commutativity of the middle square one needs to check that the boundary map $H_n(M, \partial M) \to H_{n-1}(\partial M)$ sends a fundamental class for M to a fundamental class for ∂M. We leave this as an exercise at the end of the section. □

Here is another kind of duality which generalizes the calculation of the local homology groups $H_i(M, M - \{x\}; \mathbb{Z})$:

Theorem 3.44. *If K is a compact, locally contractible subspace of a closed orientable n-manifold M, then $H_i(M, M - K; \mathbb{Z}) \approx H^{n-i}(K; \mathbb{Z})$ for all i.*

Proof: Let U be an open neighborhood of K in M. Consider the following diagram whose rows are long exact sequences of pairs:

$$\begin{array}{ccccccccc}
\cdots \longrightarrow & H_i(M-K) & \longrightarrow & H_i(M) & \longrightarrow & H_i(M, M-K) & \longrightarrow & H_{i-1}(M-K) & \longrightarrow \cdots \\
& \uparrow & & \uparrow & & \wr\uparrow & & \uparrow & \\
& H^{n-i}(M-K, U-K) & & & & H_i(U, U-K) & & H^{n-i+1}(M-K, U-K) & \\
& \wr\uparrow & & & & \uparrow & & \wr\uparrow & \\
\cdots \longrightarrow & H^{n-i}(M, U) & \longrightarrow & H^{n-i}(M) & \longrightarrow & H^{n-i}(U) & \longrightarrow & H^{n-i+1}(M, U) & \longrightarrow \cdots
\end{array}$$

$H_c^n(X,A;G)$ are defined when A is a closed subset of X, which guarantees that the inclusion $A \hookrightarrow X$ is a proper map. With these constraints the basic theory of §3.1 translates without difficulty to cohomology with compact supports.

In particular, for a locally compact CW complex X one can compute $H_c^*(X;G)$ using *finite cellular cochains*, the cellular cochains vanishing on all but finitely many cells. Namely, to compute $H_c^n(X^n, X^{n-1};G)$ using excision one first has to identify this group with $H_c^n(X^n, N(X^{n-1});G)$ where $N(X^{n-1})$ is a closed neighborhood of X^{n-1} in X^n obtained by deleting an open n-disk from the interior of each n-cell. If X is locally compact, the obvious deformation retraction of $N(X^{n-1})$ onto X^{n-1} is a proper homotopy equivalence. Hence via long exact sequences and the five-lemma we obtain isomorphisms $H_c^n(X^n, X^{n-1};G) \approx H_c^n(X^n, N(X^{n-1});G)$, and by excision the latter group can be identified with the finite cochains.

Proof of 3H.5: As noted above, we can compute $H_c^*(\tilde X;\mathbb{Z})$ using the groups $C_f^n(\tilde X;\mathbb{Z})$ of finite cellular cochains $\varphi : C_n \to \mathbb{Z}$, where $C_n = H_n(\tilde X^n, \tilde X^{n-1})$. Giving $\tilde X$ the CW structure lifting the CW structure on X, then since X is compact, finite cellular cochains are exactly homomorphisms $\varphi : C_n \to \mathbb{Z}$ such that for each cell e^n of $\tilde X$, $\varphi(\gamma e^n)$ is nonzero for only finitely many covering transformations $\gamma \in \pi$. Such a φ determines a map $\hat\varphi : C_n \to \mathbb{Z}[\pi]$ by setting $\hat\varphi(e^n) = \sum_\gamma \varphi(\gamma^{-1}e^n)\gamma$. The map $\hat\varphi$ is a $\mathbb{Z}[\pi]$-homomorphism since if we replace the summation index γ in the right side of $\varphi(\eta e^n) = \sum_\gamma \varphi(\gamma^{-1}\eta e^n)\gamma$ by $\eta\gamma$, we get $\sum_\gamma \varphi(\gamma^{-1}e^n)\eta\gamma$. The function $\varphi \mapsto \hat\varphi$ defines a homomorphism $C_f^n(\tilde X;\mathbb{Z}) \to \mathrm{Hom}_{\mathbb{Z}[\pi]}(C_n, \mathbb{Z}[\pi])$ which is injective since φ is recoverable from $\hat\varphi$ as the coefficient of $\gamma = 1$. Furthermore, this homomorphism is surjective since a $\mathbb{Z}[\pi]$-homomorphism $\psi : M \to \mathbb{Z}[\pi]$ has the form $\psi(x) = \sum_\gamma \psi_\gamma(x)\gamma$ with $\psi_\gamma \in \mathrm{Hom}_{\mathbb{Z}}(M,\mathbb{Z})$ satisfying $\psi_\gamma(x) = \psi_1(\gamma^{-1}x)$, so ψ_1 determines ψ. The isomorphisms $C_f^n(\tilde X;\mathbb{Z}) \approx \mathrm{Hom}_{\mathbb{Z}[\pi]}(C_n, \mathbb{Z}[\pi])$ are isomorphisms of cochain complexes, so the respective cohomology groups $H_c^n(\tilde X;\mathbb{Z})$ and $H^n(X;\mathbb{Z}[\pi])$ are isomorphic. □

Cup and cap product work easily with local coefficients in a bundle of rings, the latter concept being defined in the obvious way. The cap product can be used to give a version of Poincaré duality for a closed n-manifold M using coefficients in a bundle of rings E under the same assumption as with ordinary coefficients that there exists a fundamental class $[M] \in H_n(M;E)$ restricting to a generator of $H_n(M, M-\{x\};E)$ for all $x \in M$. By excision the latter group is isomorphic to the fiber ring R of E. The same proof as for ordinary coefficients then shows that $[M]\frown : H^k(M;E) \to H_{n-k}(M;E)$ is an isomorphism for all k.

Taking R to be one of the standard rings $\mathbb{Z}$, $\mathbb{Q}$, or $\mathbb{Z}_p$ does not give anything new since the only ring automorphism these rings have is the identity, so the bundle of rings E must be the product $M \times R$. To get something more interesting, suppose we take R to be the ring $\mathbb{Z}[i]$ of Gaussian integers, the complex numbers $a + bi$ with

$a, b \in \mathbb{Z}$. This has complex conjugation $a + bi \mapsto a - bi$ as a ring isomorphism. If M is nonorientable and connected we can use the homomorphism $\omega : \pi_1(M) \to \{\pm 1\}$ that defines the bundle of groups $M_{\mathbb{Z}}$ to build a bundle of rings E corresponding to the action of $\pi_1(M)$ on $\mathbb{Z}[i]$ given by $\gamma(a + bi) = a + \omega(\gamma)bi$. The homology and cohomology groups of M with coefficients in E depend only on the additive structure of $\mathbb{Z}[i]$ so they split as the direct sum of their real and imaginary parts, which are just the homology or cohomology groups with ordinary coefficients $\mathbb{Z}$ and twisted coefficients $\widetilde{\mathbb{Z}}$, respectively. The fundamental class in $H_n(M; \widetilde{\mathbb{Z}})$ constructed in Example 3H.3 can be viewed as a pure imaginary fundamental class $[M] \in H_n(M; E)$. Since cap product with $[M]$ interchanges real and imaginary parts, we obtain:

Theorem 3H.6. *If M is a nonorientable closed connected n-manifold then cap product with the pure imaginary fundamental class $[M]$ gives isomorphisms $H^k(M; \mathbb{Z}) \approx H_{n-k}(M; \widetilde{\mathbb{Z}})$ and $H^k(M; \widetilde{\mathbb{Z}}) \approx H_{n-k}(M; \mathbb{Z})$.* □

More generally this holds with $\mathbb{Z}$ replaced by other rings such as $\mathbb{Q}$ or $\mathbb{Z}_p$. There is also a version for noncompact manifolds using cohomology with compact supports.

Exercises

1. Compute $H_*(S^1; E)$ and $H^*(S^1; E)$ for $E \to S^1$ the nontrivial bundle with fiber $\mathbb{Z}$.

2. Compute the homology groups with local coefficients $H_n(M; M_{\mathbb{Z}})$ for a closed nonorientable surface M.

3. Let $\mathcal{B}(X; G)$ be the set of isomorphism classes of bundles of groups $E \to X$ with fiber G, and let $E_0 \to B\mathrm{Aut}(G)$ be the bundle corresponding to the 'identity' action $\rho : \mathrm{Aut}(G) \to \mathrm{Aut}(G)$. Show that the map $[X, B\mathrm{Aut}(G)] \to \mathcal{B}(X, G)$, $[f] \mapsto f^*(E_0)$, is a bijection if X is a CW complex, where $[X, Y]$ denotes the set of homotopy classes of maps $X \to Y$.

4. Show that if finite connected CW complexes X and Y are homotopy equivalent, then their universal covers $\widetilde{X}$ and $\widetilde{Y}$ are proper homotopy equivalent.

5. If X is a finite connected graph with $\pi_1(X)$ free on $g > 0$ generators, show that $H^n(X; \mathbb{Z}[\pi_1 X])$ is zero unless $n = 1$, when it is $\mathbb{Z}$ when $g = 1$ and the direct sum of a countably infinite number of $\mathbb{Z}$'s when $g > 1$. [Use Proposition 3H.5 and compute $H^n_c(\widetilde{X})$ as $\varinjlim H^n(\widetilde{X}, \widetilde{X} - T_i)$ for a suitable sequence of finite subtrees $T_1 \subset T_2 \subset \cdots$ of $\widetilde{X}$ with $\bigcup_i T_i = \widetilde{X}$.]

6. Show that homology groups $H_n^{\ell f}(X; G)$ can be defined using **locally finite** chains, which are formal sums $\sum_\sigma g_\sigma \sigma$ of singular simplices $\sigma : \Delta^n \to X$ with coefficients $g_\sigma \in G$, such that each $x \in X$ has a neighborhood meeting the images of only finitely many σ's with $g_\sigma \neq 0$. Develop this version of homology far enough to show that for a finite-dimensional locally compact CW complex X, $H_n^{\ell f}(X; G)$ can be computed using infinite cellular chains $\sum_\alpha g_\alpha e^n_\alpha$.